AI Technicians Research Project

Developing Rapid Occupational Training Methods for a Competitive AI Workforce

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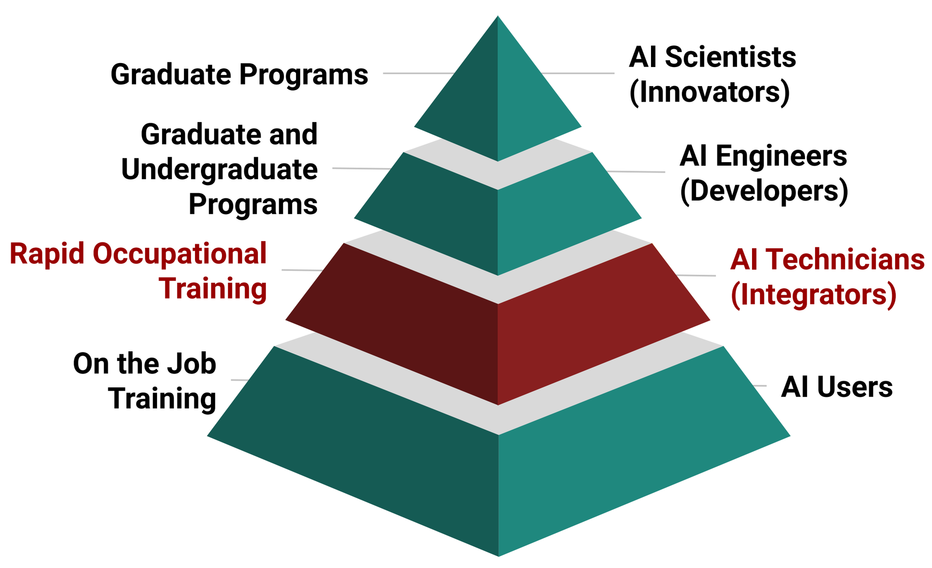
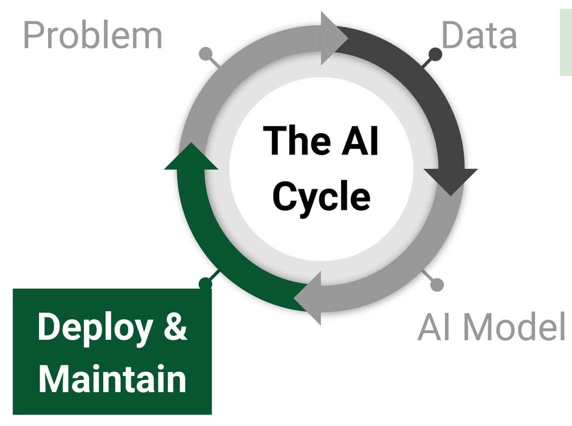
The AI/ML Workforce

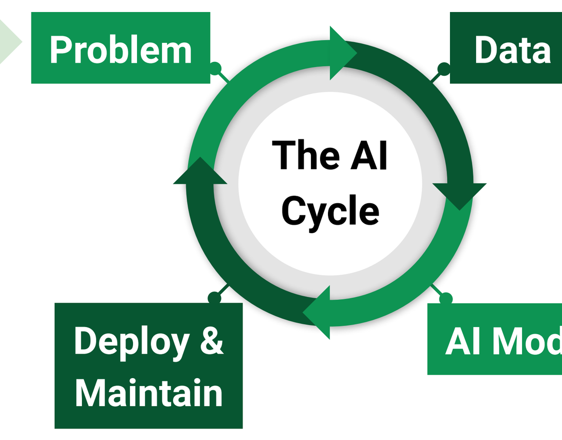




The Role of AI Technicians in the US Army

2020
- Provision & maintain the infrastructure
- Deploy & monitor the solutions

2023
- Understand emerging problems
- Data manipulation and transformation
- Support the development of AI solutions
- Maintain the AI solutions



AI Technicians Research Instrumentation

Sail() Platform - Learning science research as the first class citizen
- Well-established research-validated methods
- Sophisticated logging pipeline
- Real-time monitoring and learning analytics

In Vivo Experiments
- Curriculum A vs B
- Training method A vs B
- Tool A vs B

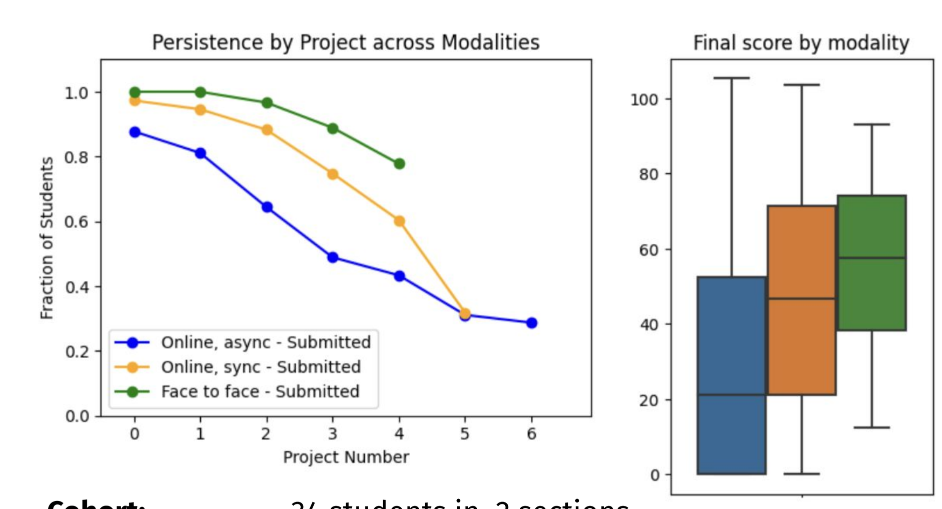
Student Surveys
- Background, attitudes
- Self-efficacy
- STEM identity

Pre- and post- knowledge and skills tests

Focus Groups
- Trainees
- Supervisors, AI2C leadership

Effective Methods for Rapid Occupational Training

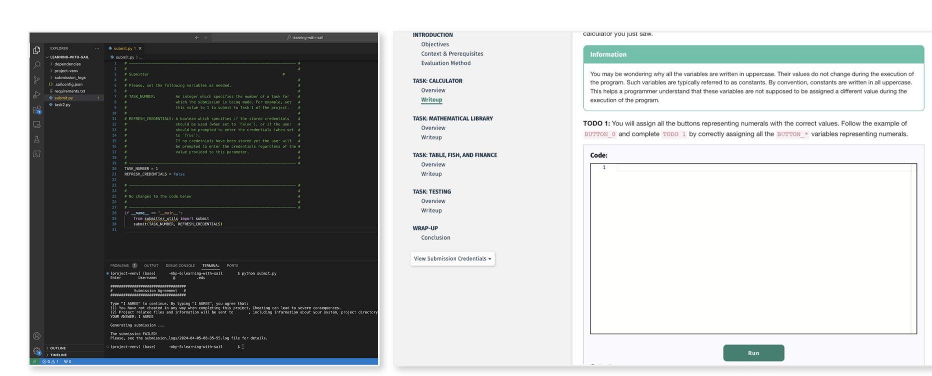
Learning Modality and Persistence

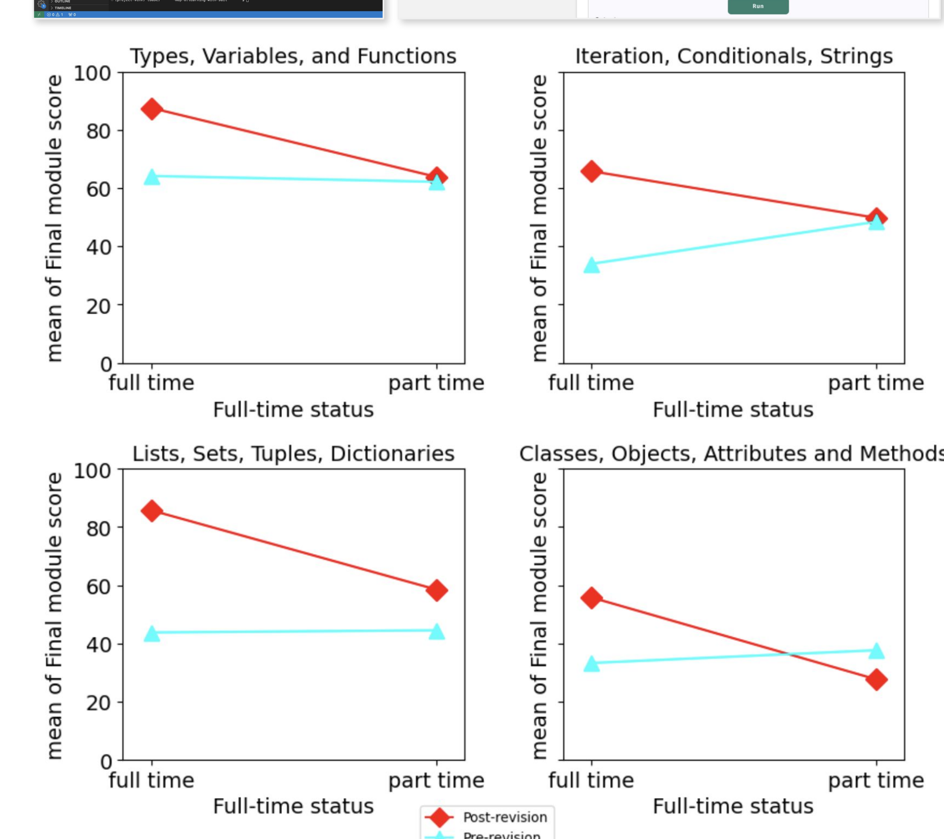


Cohort: 34 students in 2 sections
In-person: 140 students in 11 sections
Online (Sync): 205 students in 10 sections
Online (Async): 223 students in 14 sections

Bogart, C., An, M., Keylor, E., Singh, P., Savelka, J. and Sakr, M., 2024, March. What Factors Influence Persistence in Project-based Programming Courses at Community Colleges?. In Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1 (pp. 116-122).

Real vs Simplified Coding Environment



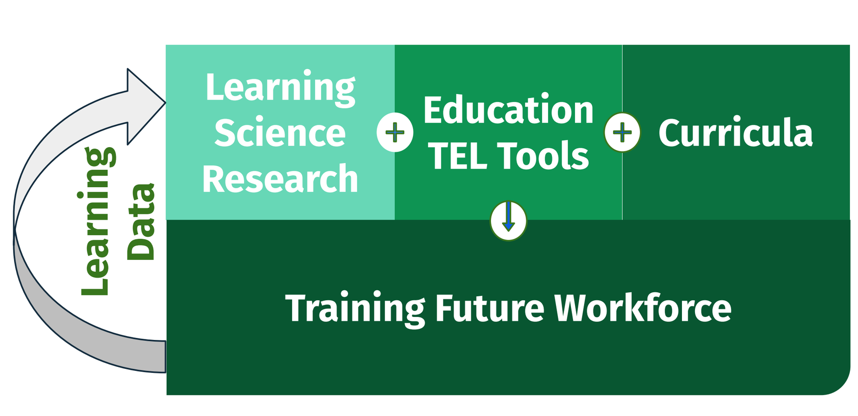


Nguyen, H., Bogart, C., Savelka, J., Zhang, Y, and Sakr, M. Examining the Trade-offs between Simplified and Realistic Coding Environments in an Introductory Python Programming Class. (in review)

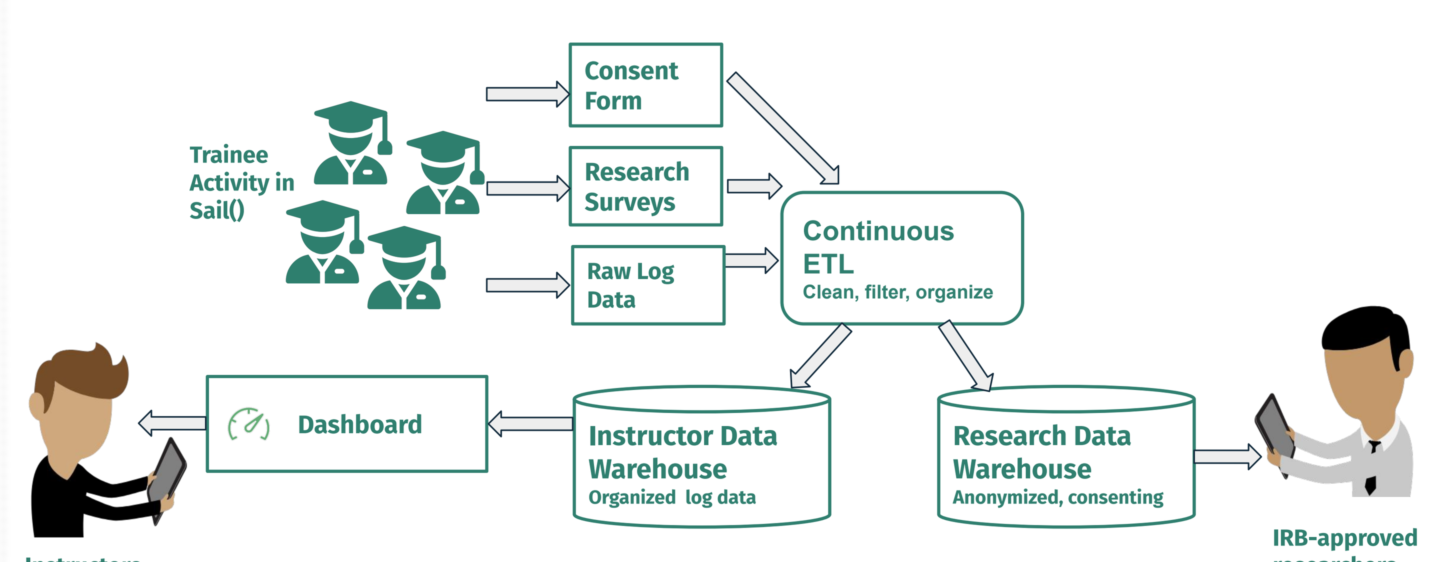
The Training Curricula Evolution

Year	Program	Trainees	Weeks	Curriculum
2020	AI Cloud Technicians	12	16	1. Cloud Admin 2. Cloud DevOps 3. Principles of Computing 4. Cloud Developer
2021	AI Cloud Technicians	15	16	1. Cloud Admin 2. Prog. with Python 3. Data Engineer 4. Cloud DevOps 5. AI Practitioner
2022	AI Cloud Technicians	18	16	1. Cloud - Administration - DevOps 2. Data and AI - Programming - Data Engineering - AI/ML
2023	AI Technicians	14	16+16	Fall: 1. Cloud Adm. 2. Programming 3. DevOps; Spring: 4. Cloud Native 5. Data Engineering 6. AI Practitioner 7. AI2C Capstone

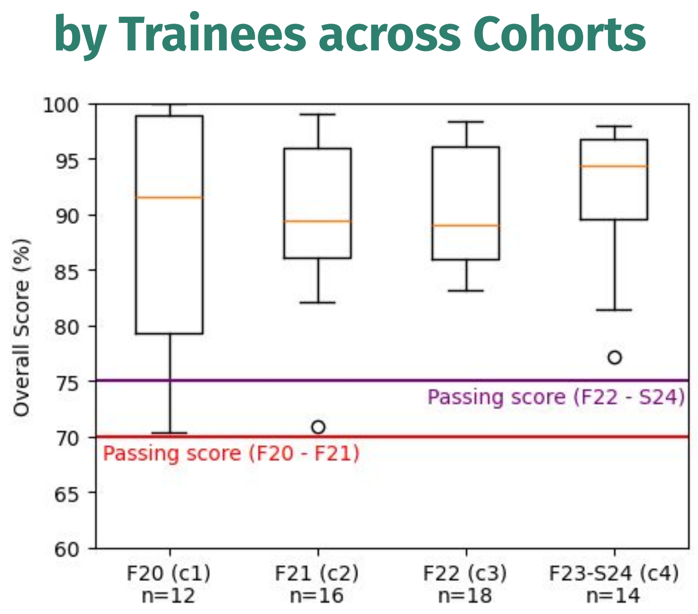
Sail() Platform - An Ecosystem



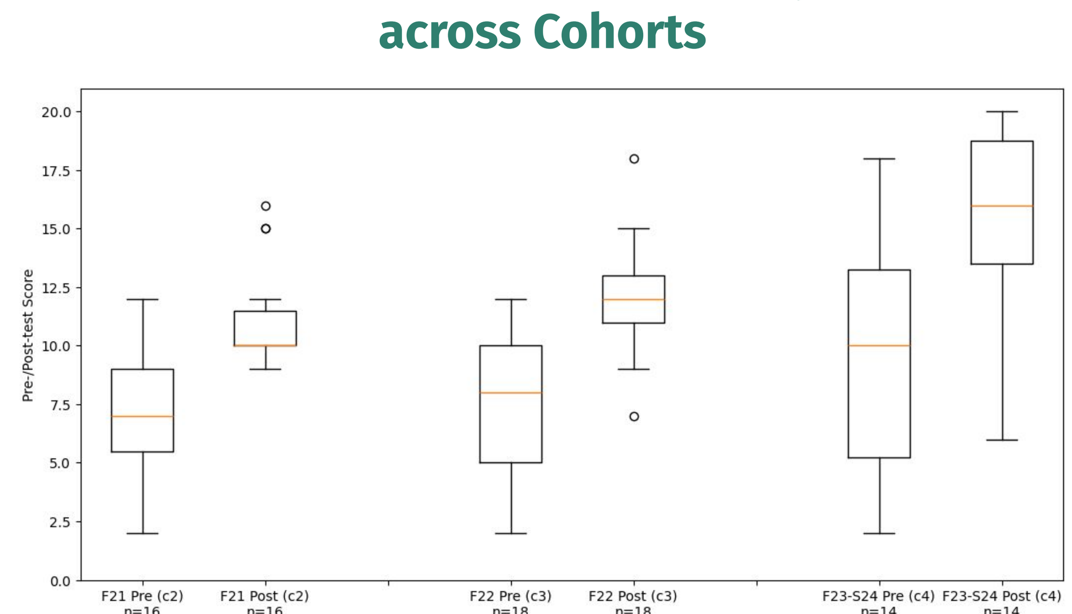
Automated Research Infrastructure of Sail()



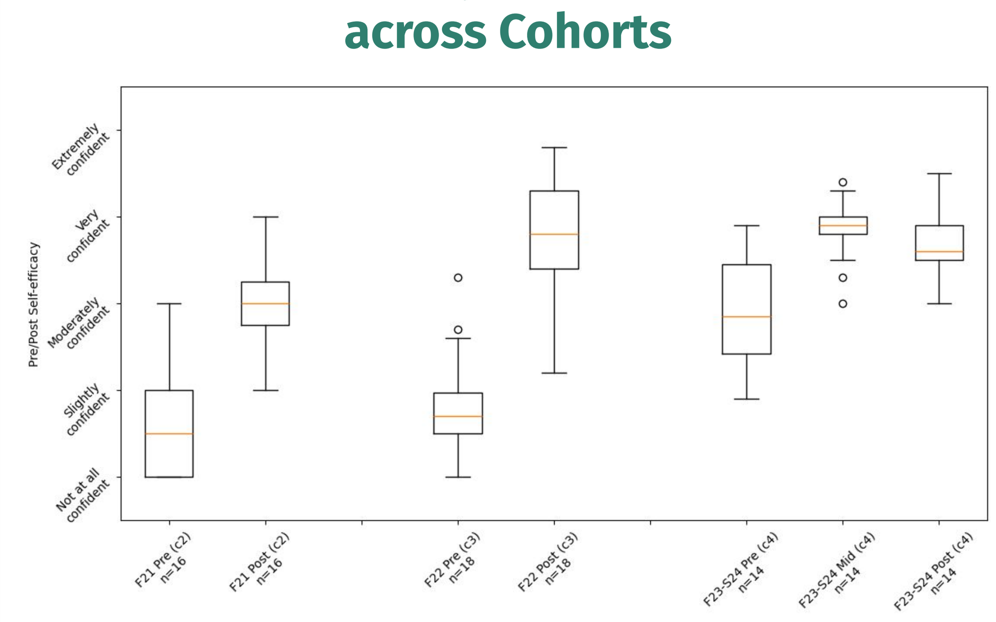
Overall Scores Achieved by Trainees across Cohorts



Pre-/Post-test Scores Achieved by Trainees across Cohorts



Self-efficacy in the Domain across Cohorts



Key Takeaways

Limited workforce for future jobs

Traditional AI education poorly reflects the need for **technicians** and **operators**

Rapid occupational training and **on the job training** appear as viable approaches

The **AI Technicians research project** continues to develop and evaluate effective methods to train the workforce of the future

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